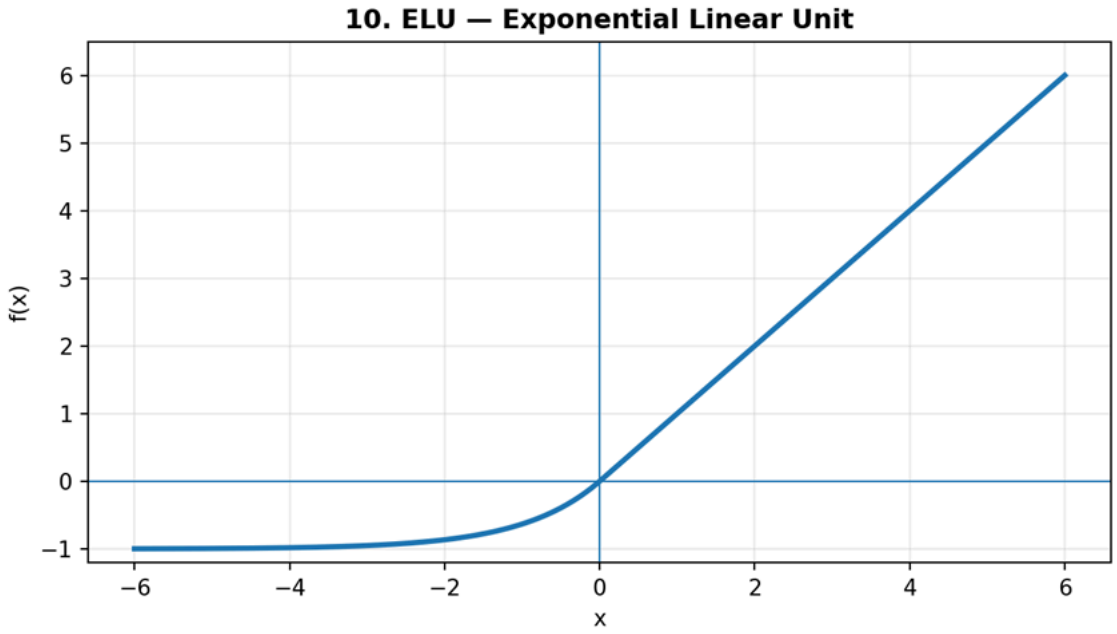
Activation Functions in Neural Networks (Cont)

ELU — Exponential Linear Unit

$$\mathrm{ELU}(x) = x \text{ if } x > 0; \quad \alpha(e^x - 1) \text{ if } x \leq 0$$

ELU — Exponential Linear Unit mathematical expression.



Characteristics

- Linear positive branch.
- Smooth exponential negative branch.
- Negative output is bounded below by −α.

Advantages

- Smooth negative branch.
- Can provide negative outputs.

Disadvantages

- More expensive than ReLU.
- Not always better empirically.

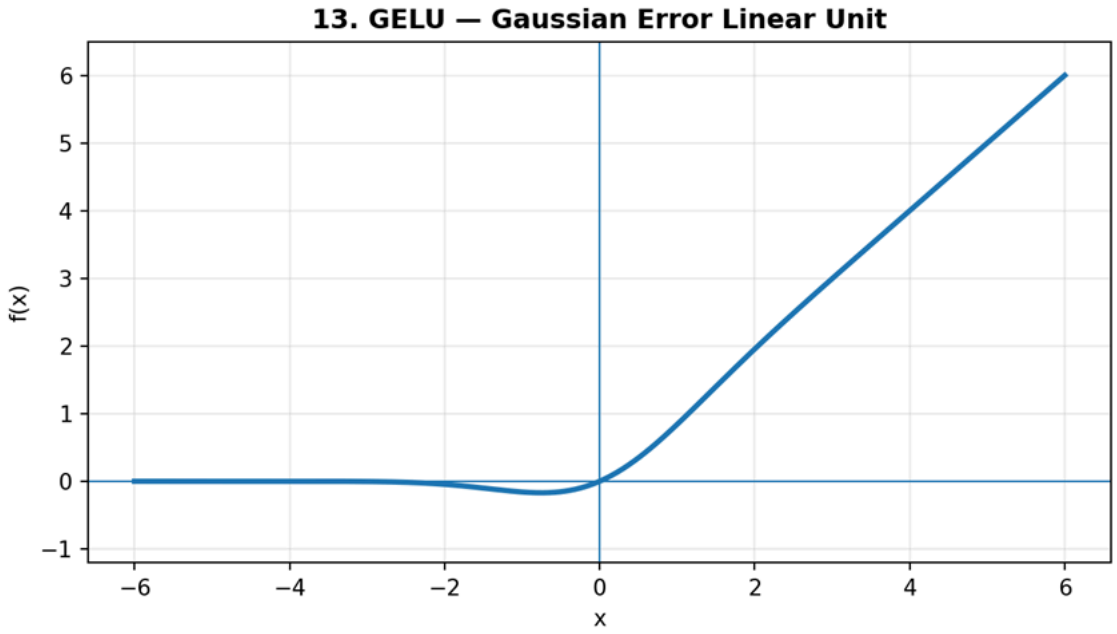
When to use

Use as a smooth ReLU alternative when negative outputs are desirable.

GELU — Gaussian Error Linear Unit

$$\text{GELU}(x) \approx 0.5x[1 + \tanh(\sqrt{(2/\pi)}(x + 0.044715x^3))]$$

GELU — Gaussian Error Linear Unit mathematical expression.



Characteristics

- Smooth and differentiable.
- Softly gates inputs rather than hard-thresholding them.
- Common in Transformer feed-forward blocks.

Advantages

- Strong empirical performance in many modern architectures.
- Smooth gating.

Disadvantages

- More computationally expensive than ReLU.
- More mathematically complex.

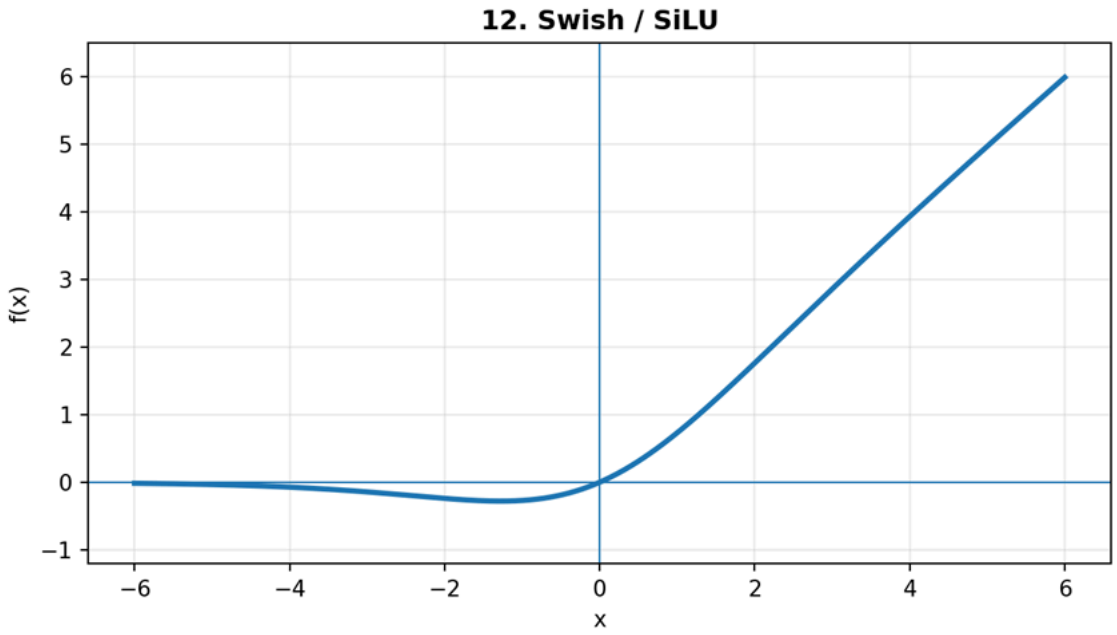
When to use

Use especially when following Transformer or other modern architectures that specify GELU.

Swish / SiLU

$$\text{Swish}(x) = x \cdot \sigma(x) = x \,/\, (1 + e^{-x})$$

Swish / SiLU mathematical expression.



Characteristics

- Smooth and non-monotonic.
- Allows a small negative region.

Advantages

- Often effective in modern deep architectures.
- Smooth optimization behavior.

Disadvantages

- More computation than ReLU.
- Not guaranteed to outperform simpler choices.

When to use

Use in modern architectures when its empirical performance and computational cost are acceptable.

Softmax

$$P(y=i) = e^{z_i} / \Sigma_j e^{z_j}$$

Softmax converts a vector of logits into a probability distribution.

Class	Logit	Illustrative probability
Cat	2.5	≈ 0.73
Dog	1.2	≈ 0.20
Horse	0.3	≈ 0.07

Characteristics

- Outputs lie between 0 and 1.
- All probabilities sum to 1.
- Applied to a vector of class logits rather than independently to hidden neurons.

Advantages

- Natural probability distribution for mutually exclusive classes.
- Pairs naturally with multi-class cross-entropy.

Disadvantages

- Not normally used in hidden layers.
- Not appropriate for independent multi-label outputs.
- Use numerically stable implementations.

When to use

Use at the output of a single-label multi-class classifier. For multi-label classification, use independent sigmoid outputs.

Choosing the Output Activation

Task	Recommended output	Reason
Binary classification	Sigmoid	One probability between 0 and 1
Single-label multi-class classification	Softmax	Probabilities form one distribution
Unrestricted regression	Linear / identity	Output can be any real number
Positive-only regression	Often Softplus or suitable transform	Guarantees positive output
Bounded regression	Scaled sigmoid/tanh	Restricts output to a known interval

Choosing a Hidden-Layer Activation

Situation	Good starting choice	Why
General deep network	ReLU	Fast, simple, strong baseline
ReLU causes dead neurons	Leaky ReLU	Keeps a negative gradient
Want learned negative slope	PReLU	Slope is trainable

Transformer-style architecture	GELU	Common architecture choice
Modern architecture using SiLU	Swish/SiLU	Smooth and effective
Bounded zero-centered state	tanh	Range is −1 to +1

Why Activation Shape Matters

Backpropagation uses derivatives. If an activation derivative becomes extremely small, gradients can shrink through layers. Sigmoid and tanh can saturate; ReLU avoids positive-side saturation but has exactly zero derivative on its negative branch.

$$\partial L/\partial w = (\partial L/\partial a) \cdot (\partial a/\partial z) \cdot (\partial z/\partial w)$$

Simplified chain-rule view of a neuron's weight gradient.

ReLU Sparsity

ReLU is called sparse because negative inputs become exactly zero. For example:

Input	−3	−1	0.5	2	−4	1
ReLU output	0	0	0.5	2	0	1

Practical Rules of Thumb

- Start with ReLU for ordinary hidden layers unless the architecture suggests otherwise.
- Binary classification output → sigmoid.
- Single-label multi-class output → softmax.
- Unrestricted regression → linear output.
- If ReLU neurons die → try Leaky ReLU or PReLU and inspect activation statistics.
- Use tanh when a bounded zero-centered state is specifically useful.

- Use GELU or Swish when the architecture calls for them or experiments show an advantage.
- Do not choose an activation only because it is newer, validate it on the actual task.